MILITARY SPECIFICATION

HOOK; BOMB SHACKLE, AERO 3A

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the manufacture, inspection and testing of one type of bomb shackle hook for use on the bomb shackle, Aero 3A, and hereinafter referred to as hook.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 The following specifications, publications and standards of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

LLL-B-631	Boxes; Fiber, Corrugated (For Domestic Shipment)
LLL-B-636	Boxes; Fiber, Solid (For Domestic Shipment)
QQ-S-624	Steel, Alloy; Bars (General Purpose)

Military

MIL-P-116	Preservation, Methods of
MIL-A-140	Adhesive Water-Resistant, Waterproof Barrier-Material
MIL-S-5000	Steel, Chrome-Nickel - Molybdenum (4340) Bars and Reforging Stock
MIL-S-5626	Steel, Chrome-Molybdenum (4140) Bars, Rods and Forging Stock (For Aircraft Applications)
MIL-A-6870(ASG)	Inspection Requirements, Magnetic Particle
MIL-P-6871	Plating, Chromium
MIL-H-6875(ASG)	Heat Treatment of Steels (Aircraft) Process
MIL-S-7108	Steel, Alloy (Manganese-Silicon - Nickel), Bars and Forging Stock
JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-120	Packaging and Packing for Overseas Shipment - Cartons, Folding, Paperboard

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Military (Cont'd)

JAN-B-121	Barrier - Materials, Greaseproof
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier - Materials, Waterproof, Flexible
JAN-P-133	Packaging and Packing for Overseas Shipment - Boxes, Set-Up, Paperboard

Navy Department

General Specification for Inspection of Material

STANDARDS

MIL-STD-129	Marking of Shipments

DRAWINGS

52A328C29	Hook, Detail, Bomb Shackle, Aero 3A

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

(When requesting specifications, publications and drawings, refer to both title and number. Copies of this specification, applicable specifications and drawings may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania, Attention: Technical Records Division.)

3. REQUIREMENTS

3.1 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials which are not covered by applicable specifications shall be of the magnaflux quality steel, and entirely suitable for the purpose intended.

3.1.1 Critical Materials - If the materials specified in this specification, applicable specifications and/or drawings for the production of the subject item are critical materials as defined by the National Production Authority, the contractor shall list such materials and notify the Procuring Service of its availability to the contractor. In cases where materials are difficult to procure, the contractor shall also prescribe which critical or less critical material can be substituted without departing, revising or changing the original design and/or functioning of the subject item. No substitution of such material shall be permitted without prior written approval of the Procuring Service.

3.1.2 Selection of Materials - Specifications and standards for all materials, parts and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143.

3.1.2.1 Weight Reduction - The contractor shall make every effort to reduce the over-all weight of the equipment required by this specification without impairing the function, operation or durability of the equipment being produced. If in the opinion of the contractor a considerable saving in weight can be realized by incorporating specific changes, it is requested that comments to that effect be submitted to the Chief, Bureau of Aeronautics, by means of a letter or by consultation. In any event, changes to equipment required by this specification, will not be permitted unless such authorization is granted by the Chief, Bureau of Aeronautics in writing.

3.2 Dimensions - The hook shall be manufactured to conform to the design and dimensions shown in Bureau of Aeronautics Drawing 52A328C29.

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3.3 Finish - The hook shall be chromium-plated in accordance with Specification MIL-P-6871, Type II. The thickness of the plating shall be 0.002 inch minimum. Plating in the holes is not required.

3.3.1 Wear resistance - The wear resistance of the lug-bearing surface after plating shall be equivalent to that specified in the standard Specification MIL-P-6871, Type II specimen.

3.4 Hardness - The hook shall be hardened sufficiently to meet the load requirements. This will require a Rockwell C45 to C48 hardness rating. (See Section 6 for recommended heat treatment.) Hardness measurements shall be made after heat treatment. The lug-bearing surface shall have the same resistance to brinelling and galling as a polished steel specimen hardened to Rockwell C45 minimum and plated in accordance with paragraph 3.3.

3.5 Performance - The hook shall be capable of withstanding the loads specified in paragraphs 3.5.1, 3.5.2, and 3.5.3 when tested in accordance with paragraph 4.5, and shall perform satisfactorily as specified in Section 4.

3.5.1 Static load - The hook shall withstand a static down load of 25,500 pounds for 60 seconds minimum, performed in conjunction with the ultimate strength requirement of paragraph 3.5.3.

3.5.2 Dynamic load - The hook shall withstand a minimum of 150,000 cycles at a rate of 1,800 cycles per minute, dynamically loaded during each cycle beginning with a load of 2,250 pounds, increasing to 7,750 pounds, and reducing to 2,250 pounds. The load-time curve shall be equivalent to that produced by the Sonntag fatigue testing machine.

3.5.3 Ultimate strength - The ultimate strength of the hook shall be greater than 25,500 pounds vertically down.

3.6 Hook deformation after prestressing - The maximum deformation of the hook after prestressing, if applied as recommended in paragraph 6.3.6, shall not cause the dimension of the lower pivot hole to exceed 0.380 inch along all axes.

3.7 Identification of product - Each hook manufactured shall be stamped as designated in Bureau of Aeronautics Drawing 52A328C29.

3.8 Workmanship - The workmanship shall be of the highest quality employed in the production of the type of hooks covered by the specification and, in all respects, shall be satisfactory as determined by the Government Inspector. The manufacturing processes used shall be adequately controlled to produce hooks which will meet the requirements herein. Suggested heat treating and manufacturing processes for certain types of steels are presented in Section 6 as an aid to the manufacturer.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Classification of tests - The inspection and testing of hooks shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the production of the items after the award of contract, to determine that the production meets the requirements of this specification and to establish the required prestress load, if applied as recommended.

(b) Inspection tests: Inspection tests are those tests accomplished on bomb shackle hooks manufactured and submitted for acceptance under contract.

4.2 Inspecting and testing agencies - The hooks together with materials and manufacturing processes used in their construction shall be inspected in general accordance with Navy Department Specification for the Inspection of Materials by authorized Government Inspections at the contractor's plant or elsewhere as directed by the procuring agency. Preproduction samples shall be tested in accordance with the tests specified herein by the Government activity or other testing laboratories to be designated by the procuring agency; the contractor shall not conduct tests, other than Individual tests.

4.3 Preproduction tests -

4.3.1 Sampling instructions - The first 25 hooks manufactured by any one manufacturer or under each contract (as determined by the procuring agency) and fabricated by quantity production

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methods and tools shall be the preproduction samples. The preproduction samples shall be delivered for inspection and testing as directed by the procuring agency. Certification from the cognizant Government Inspector that the materials used in the samples are in accordance with requirements given herein (see paragraph 3.1) shall be forwarded with the samples. The preproduction samples shall be labeled "No. 1, No. 2, No. 3", etc., through "No. 25" respectively.

4.3.2 Tests - Each preproduction sample shall be tested in accordance with the tests specified below and described under "Test Methods," and in the order shown, except that only one-third of the preproduction samples shall be subjected to the Ultimate Load test:

(a) Examination of product (paragraph 4.5.1)

(b) Static load test (paragraph 4.5.2)

(c) Ultimate strength test (on 1/3 of the samples) (paragraph 4.5.3)

(d) Hole test (paragraph 4.5.4)

(e) Dynamic load test (paragraph 4.5.5)

4.4 Inspection tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.4.1 Individual tests - Each hook shall be subjected to the following tests, described under "Test Methods", paragraph 4.5.

(a) Examination of product

(b) Hole test

4.4.2 Sampling tests - Hooks shall be selected at random from each lot of 100 hooks manufactured and shall be submitted as the sampling test samples. The samples shall be suitably labeled to show lot numbers. The following table indicates the number of samples to be submitted per production lot.

Quantity of hooks in production lot	Quantity of samples to be selected
100 or less	3
101 to 200	5
201 to 300	7

For sampling purposes, lots of over 300 hooks shall be divided into smaller lots in accordance with the above table.

4.4.2.1 Samples selected shall be subjected, in addition to the Individual tests, to the following tests as described under "Test Methods", (paragraph 4.5):

(a) Static load test

(b) Ultimate load test

(c) Dynamic load test

4.4.3 Rejection and retest -

4.4.3.1 Rejection of preproduction samples - The failure of any of the 25 preproduction samples to pass the required tests, (see paragraph 4.3.2), shall be cause for tentative rejection of all the preproduction samples. No deliveries of production hooks shall be accepted until the preproduction samples satisfactorily complete the required tests and all corrections and changes made to the preproduction samples are incorporated in production.

4.4.3.1.1 Correction of defects in preproduction samples - Any defects constituting noncompliance with the requirements of this specification, which are disclosed during inspection and

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testing of the preproduction samples will be corrected by the contractor at his expense. Such corrective measures as may be found necessary shall likewise be incorporated in the production of all subsequent hooks submitted for acceptance under the contract.

4.4.3.1.2 Retest of preproduction samples - Preproduction samples which have been rejected and resubmitted for test shall be tested in accordance with one or more of the test methods described under paragraph 4.5 as determined by the testing activity or cognizant Government Inspector.

4.4.3.2 Rejection of individual unit - The failure of any hook to pass the inspection and tests required in paragraph 4.4.1 shall cause its tentative rejection, subject to retest as hereinafter provided. (See paragraph 4.4.3.2.1.)

4.4.3.2.1 Retest of individual units - Any hook which fails to pass the Inspection tests as required in paragraph 4.4.1 may be resubmitted upon the request of the contractor for individual inspection and test, provided a statement showing the corrective measures taken is furnished to the Inspector at the time of such resubmission. Upon reinspection and retest, failure to meet the requirements outlined in paragraph 4.4.1 shall be cause for final rejection of the unit.

4.4.3.3 Rejection of sampling test samples - The failure of any production lot sample to pass the inspections and tests required in paragraph 4.4.2 shall cause tentative rejection of all hooks in the lot represented by that sample, subject to retest as hereinafter provided. (See paragraphs 4.4.3.3.1 and 4.4.3.3.2.)

4.4.3.3.1 First retest, sampling test samples - In the event any sample representing a production lot fails to pass the inspection and tests required in paragraph 4.4.2 the contractor shall have the option of reworking the entire affected lot and resubmitting to retest twice as many samples, selected at random by the Government Inspector, as originally submitted for the lot. (See paragraph 4.4.1.) Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected again.

4.4.3.3.2 Second retest, sampling test samples - If, in the contractor's opinion, the failure of the retest samples is due to a remediable defect, the contractor may, at the discretion of the Government Inspector, be offered the opportunity of reworking the entire affected lot and resubmitting for a second retest, three times as many samples, selected at random, as originally submitted for the lot (see paragraph 4.4.2). Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be finally rejected.

4.4.3.4 Inspection and testing subsequent to delivery - Any or all hooks submitted subsequent to the first 25 preproduction samples (see paragraph 4.3) may be subject to inspection and testing to determine their conformance with any or all requirements and tests of this specification after delivery to the Government. Any hook found to be defective due to noncompliance with the requirements of this specification and applicable specifications and drawings may be rejected after delivery.

4.5 Test methods -

4.5.1 Examination of product - Each hook shall be thoroughly examined to insure that the dimensions, workmanship, hardness, and finish are in compliance with the intent of this specification. The hook shall be subjected to magnetic particle inspection in accordance with Specification MIL-I-6870. Special attention shall be given to the throat area and lower-hole area of the hooks. Both longitudinal and circular magnetization methods shall be used. A current of 2,000 amperes minimum shall be applied for the magnetization method using a copper bar at:

(a) 5/16 hole

(b) 3/8 hole

(c) throat

A 2,000-amperes current shall also be used for the longitudinal magnetization method, and the current shall be so passed that the flux is parallel to the length of the hook. Magnetic particle inspection should be performed prior to chromium plating and also after chromium plating and prestressing. Should prestressing not be necessary to result in a satisfactory hook as required by Section 3, then magnetic particle inspection should be accomplished prior to chrome plating and as the final operation after chromium plating and subsequent operations are performed.

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4.5.2 Static-load test - The hook shall be attached to a suitable fixture in a position such that (a) the lug-bearing surface of the hook is horizontal, (b) the direction of the applied test load is normal to the lug-bearing surface, 7/16 inch from the vertical centerline of the lower pivot hole of the hook, and (c) the restraining forces are taken horizontally at the upper hole of the hook. The fixture on which the hook is tested shall be universally mounted at both ends. The fixture shall be generally in accordance with the arrangement shown in figure 1. A static load of 25,500 pounds shall be hung on the hook for a period of 60 seconds minimum.

4.5.3 Ultimate-load test - The hook shall be attached to the fixture and loaded in the same manner as required for the static-load test described in paragraph 4.5.2. The hook shall be subjected to a down load of 25,500 pounds increased to the breaking load. The ultimate load for each hook tested shall be recorded. This test shall be conducted immediately following the static-load test without removing the hook from the test fixture.

4.5.4 Hole test - A gage shall be applied to the lower pivot hole. Hooks which have deformed beyond the limit required in paragraph 3.6 shall be rejected without recourse to the retest provisions prescribed in paragraph 4.4.3. The gage shall conform to the requirements of paragraph 3.6.

4.5.5 Dynamic-load test - The hook, to be subjected to a "pull-pull" type of test, shall be attached to a suitable fixture similar to figure 2 , in such a position that:

(a) the lug-bearing surface of the hook is horizontal

(b) the direction of the applied test loads is normal to the lug-bearing surface, 7/16 inch from the vertical centerline of the lower pivot hole of the hook, and,

(c) the restraining forces are taken horizontally at the upper hole of the hook.

The hook shall be subjected to the dynamic loading of paragraph 3.5.2.

5. PREPARATION FOR DELIVERY

5.1 Application - The requirements specified herein apply only to direct purchases by or direct shipments to the government.

5.2 Preservation - Each hook shall be preserved in accordance with the requirement of Method 1A of Specification MIL-P-116, without the use of preservative compound. Each preserved hook shall be placed in a carton or box conforming to Specification JAN-P-120 or Specification JAN-P-133 unless a container of equivalent or greater rigidity was used in conjunction with the preservation method.

5.3 Packing - Unless otherwise specified, all hooks shall be packaged for domestic shipment. Shipping containers shall contain the same number of hooks, shall be uniform in size, and snugly packed.

5.3.1 Domestic packing - Unless otherwise specified, 70 preserved hooks shall be placed in a shipping container conforming to Specification LLL-B-631 or LLL-B-636.

5.3.2 Overseas packing - When specified in the contract, four of the fiberboard packages, each containing 70 preserved hooks, shall be packed in cleated plywood, or nailed wood boxes conforming to Specifications JAN-P-104 to JAN-P-106. Boxes shall be furnished with a sealed case-liner fabricated from case-liner material, conforming to Specification JAN-P-125, sealed with adhesive, conforming to Specification MIL-A-140.

5.4 Marking of shipments - Interior packages, containing 70 hooks, and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Hook, Bomb Shackle, Aero 3A, Specification MIL-H-17913(Aer).

6. NOTES

6.1 Compliance with specifications - Strict compliance with the requirements and intent of this specification and applicable referenced specifications is required unless express authorization for deviation is granted by the procuring agency. An equitable adjustment in accordance with the applicable contract provisions will be made in the price of the articles to be delivered to reflect any change so authorized by the procuring agency.

6.2 Information required with bids - Each bid shall be accompanied by a statement as to whether or not any exceptions are being taken to the requirements shown in this specification. If

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exceptions are being taken or deviations are anticipated, alternative proposals shall be indicated in the statement and submitted therewith.

6.3 Processing information - The following information is presented as a recommended guide to the manufacturer. This information is based on test and experimentations conducted under direction of the Government for the purpose of determining a satisfactory manufacturing process for the hooks. The contractor may use this information at his discretion, but compliance with the procedures set forth below shall not necessarily insure acceptance of the finished hooks. Acceptance of the hooks will be contingent upon satisfactory performance and test results required in Sections 3 and 4.

6.3.1 Forging - The forging-furnace conditions shall be such as to give a minimum of decarburization. In the first heats obtained, the extent of decarburization shall be determined in order that adequate allowances may be made for completely grinding off all decarburization resulting both from forge and heat treatment. Adequate stock shall be retained on the forging blank to permit the removal of at least 0.01 inch of stock from all surfaces.

6.3.2 Heat treatment - The forgings shall be heat-treated following the forging operation. The heat treatment shall be in accordance with Specification MIL-H-6875 except as modified and amplified herein.

6.3.2.1 Furnace - The furnace or furnaces used for normalizing, annealing, and quenching preferably shall be of the continuous type with an adjustable speed conveyor and shall be operated through automatic temperature controllers to maintain the temperature of all points in the working zone within the required temperature ranges. The furnace used for tempering shall be adequate to heat uniformly all parts of the furnace load, including the trays or racks by forced convection of the furnace atmosphere. All furnaces shall be the controlled-atmosphere type in order to prevent decarburization.

6.3.2.2 Loading - During normalizing, annealing, quenching and tempering, the parts shall be spaced adequately on the conveyor tray or rack to allow uniform heating and quenching. The parts shall be loaded in a manner so as to permit uniform air circulation around all surfaces of the hook and to permit uniform heating of all hooks in the load.

6.3.2.3 Heating - The time at heat required for the various operations shall begin when the forgings and furnace work zone have reached the required temperature. The temperature of all points in the work zone shall be within ±10°F of the temperature required for the parts, and should not fluctuate more than 10°F throughout the heating operation.

6.3.2.4 Normalizing - Following the forging operation, the parts shall be normalized.

6.3.2.5 Annealing - Following the normalizing operation, the parts shall be annealed in accordance with Specification MIL-H-6875.

6.3.2.6 Hardening - Following the annealing operation, the parts shall be hardened in accordance with paragraph 3.4.

6.3.2.7 Tempering - Immediately following the quenching process, the parts shall be tempered. Experience with certain engineering steels indicate that for these steels the parts should be cooled after tempering by oil quenching; also, the parts should be retempered to the same temperature, and oil-quenched. If hardness measurements are taken on any hook, the impression shall be made in the shaded area indicated on Bureau of Aeronautics drawing 52A328C29.

6.3.3 Scale removal - Scale shall be removed following the tempering process. Removal of scale shall be performed by sand blasting or other mechanical method unless pickling or electrolytic methods are specifically approved by the procuring agency.

6.3.4 Machining - Care shall be exercised during the machining of the hooks to prevent irregularities and tool marks on the hooks, especially in the throat area of the hooks under the 3/8-inch hole along the 0.312- and 0.187-inch radii, and also inside the 3/8-inch hole. Excessive tool marks on the hooks shall be cause for rejection.

6.3.5 Grinding - Grinding practice shall be used for tool-steel grinding. A suitable coolant shall be used during grinding. All grinding shall be longitudinal to the hook; in no case across the hook. Grinding in the throat area shall follow the curve of the hook on all surfaces. Checks and cracks due to grinding shall be cause for rejection of the hook.

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6.6.3 Prestressing - Experience has indicated that prestress is essential to obtain the required fatigue life. In order to secure a substantial increase in fatigue life, it is recommended that all hooks be prestressed by cold stretching, as illustrated in figure 3. This prestressing operation benefits the hook by cold working the outer surface of the metal and also by placing the surface fibers under residual compressive stress. The prestress operation, when used, shall be applied to the hook for a minimum of 30 seconds and shall be the final operation in the manufacture of the hooks. In the prestressing operation, the hook is to be held so that its lug-bearing surface shall be contacted over 1/4 inches of the outermost portion of the hook surface as shown on figure 3. If the contractor wishes to substitute another method of surface treating the hooks to increase their fatigue strength, prior approval shall be obtained from the procuring agency. The prestress load shall be determined by the Government during preproduction tests, as specified in paragraph 4.3.2.

6.3.6.1 Prestress load - The required prestress load shall be determined by the Naval Air Development Center, Johnsville, Pennsylvania, during the Government conducted tests of the preproduction samples. In general, this load shall be approximately 75 per cent of the ultimate strength of the hook, subject to the deformation requirement described in paragraph 3.6. All hooks subsequent to the preproduction samples shall be prestressed by the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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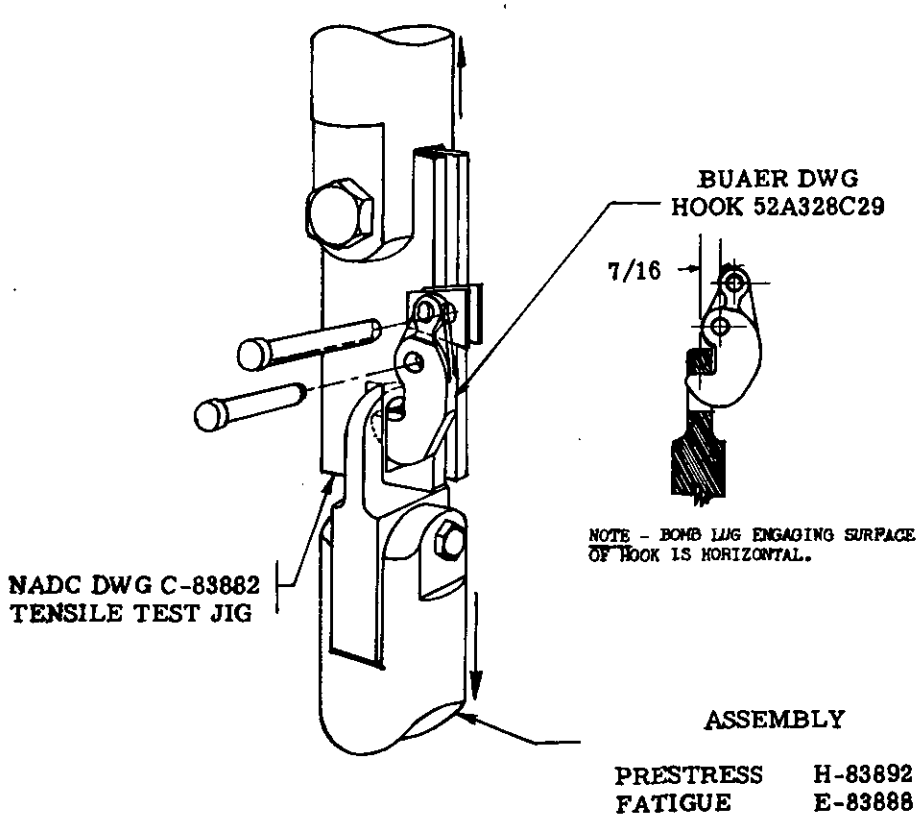


FIGURE 1

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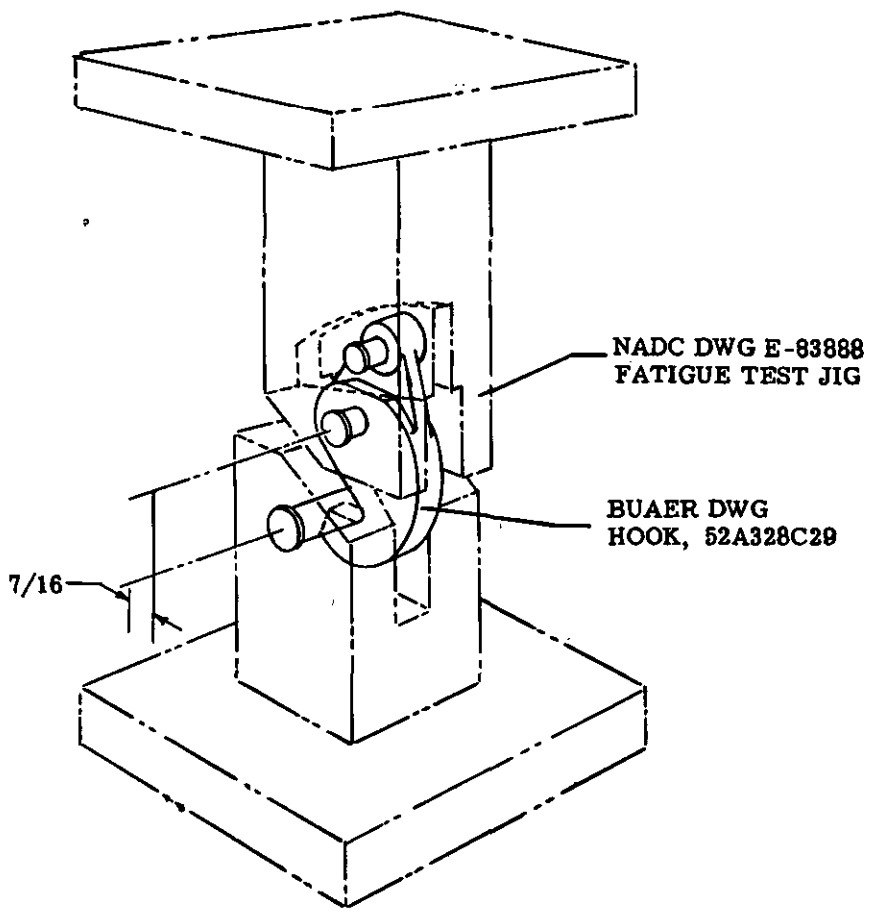


FIGURE 2

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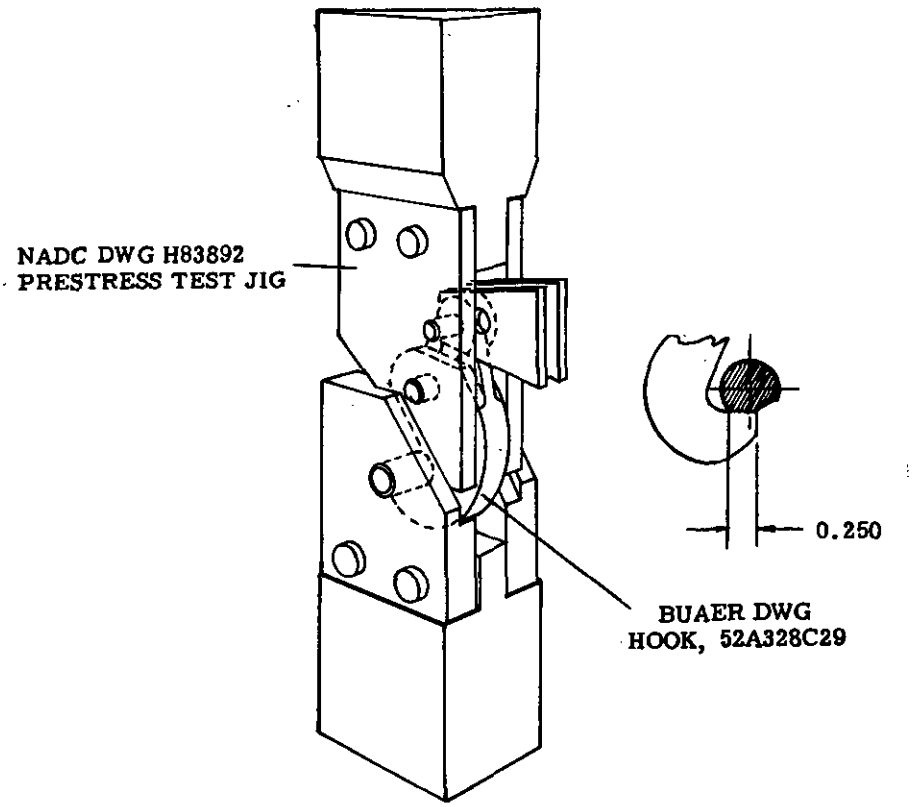


FIGURE 3

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